Jayaditya Vetsa

Leander, TX | (737) 230-0202 | jayadityavetsa@utexas.edu | Portfolio | GitHub | LinkedIn

EDUCATION

The University of Texas at Austin May 2030
Bachelor of Science in Computer Science, Robotics Honors Program *Austin, TX*
Minor in Robotics
Texas Aerial Robotics: Raytheon AVC Software Team
Texas Humanoids: State Estimation & Perception Engineer

EXPERIENCE

Undergraduate Research Assistant May 2026 - Present
The University of Texas at Dallas | Advisor: Prof. Vibhav Gogate *Richardson, TX*

- Built a **3D robot-motion evaluation pipeline** with **Python, PyTorch, Point4D, and CAD-based kinematics**, benchmarking **11,264 persistent surface points per video** to identify tracking failures and assess training-data quality.
- Reduced average position error **21%** and increased points tracked within **5 cm from 36% to 51%** across **six synthetic clips** through vision-guided tracking refinement using **PyTorch and OpenCV**.
- Reduced held-out robot-pose error from **19.99 to 2.01 degrees MAE** across **793 frames from 12 unseen episodes**.
- Accelerated **LeRobot v3 RGB conversion 76.9x to 2,383.7 frames/s**, preserving identical output for research workflows.

Machine Learning Intern Summers 2023 - 2025
Calimatic Ed Tech *Atlanta, GA / Remote*

- Built and deployed a **Python RAG service** on Azure with **1,000+ successful executions** using the OpenAI API
- Integrated **LangChain, LlamaIndex, and Pinecone** vector search to support retrieval and prompt-engineering workflows
- Shipped 5+ n8n automations for lead management, agile testing, task tracking, web operations, and improved efficiency by 70%

SOFTWARE PROJECTS

SplineFlow-Panda | *Python, MuJoCo, B-Splines, Control, Experiment Design* | GitHub Aug. 2026

- Built a reproducible **MuJoCo benchmark** and **Streamlit visualization** pipeline for comparing 100 Hz control policies
- Completed both paired **2x-speed trials in 5.0 s with B-splines** while action chunks failed the 2 cm waypoint criterion
- Processed and visualized **simulation telemetry**: tracking error, acceleration, jerk, contacts, and actuator saturation

AI-GP Autonomous Drone Stack | *Python, OpenCV, PnP, MAVLink, Sensor Fusion* | GitHub Aug. 2026

- Integrated camera-based PnP sensing with **Kalman-filtered IMU data** for onboard state estimation
- Engineered **MAVLink control modes** and state logic for detection, alignment, and autonomous navigation
- Logged telemetry and **implemented recovery logic** for signal dropout, localization failures, and collision events during runs

PUF Authentication: Adversarial Bit-Flip Attacks | *Python, Adversarial ML | IEEE HOST Submission* Jun. 2026 - Present

- Designed a **ridge-linear surrogate** to rank candidate bit flips for targeted PUF forgery attacks, validating on multiple seeds
- Improved targeted-forgery performance **212x over random flipping**, reaching a 20.03% peak rate at a 12-bit budget
- Trained a logistic rejection gate that reduced forgery success below **0.51%** while preserving **99.094% legitimate accuracy**
- **Evaluated attack-defense trade-offs** across perturbation budgets, visibility settings, and created a Streamlit dashboard

Robot SphereShield | *Python, PyTorch, PyBullet, URDF, SLSQP* | GitHub Jul. 2026

- Built a deterministic **61-sphere PyTorch/URDF system model** and benchmarked it on a 20,000-pose holdout
- Validated zero false-safe predictions across **2,479 collisions** and raised holdout accuracy from **80.71% to 94.28%**
- Optimized two colliding trajectories to **0/301 collisions** and independently validated both in PyBullet

PatchProof | *TypeScript, Python, Node.js, GitHub Actions* | GitHub | npm Jun. - Jul. 2026

- Published a **CI/CD developer tool** and GitHub Action for deterministic regression testing across **pytest, Jest, and Vitest**
- Validated compatibility across Python 3.10-3.14, Node.js 22/24, and **operating systems: linux, mac, and windows**
- Added isolated work trees, execution-consent gates, and privacy-redacted regression reports to the action dashboard

TECHNICAL SKILLS

Languages: Python, Java, TypeScript, JavaScript, C++, Swift, SQL, R
AI/ML: PyTorch, TensorFlow, scikit-learn, surrogate modeling, learned policies, model evaluation
Tools: Git/GitHub Actions, Azure, Docker, FastAPI, Node.js, OpenCV, MAVLink, Streamlit, n8n
Modeling & Controls: MuJoCo, Isaac Sim, PyBullet, URDF, SLSQP, sensor fusion, Kalman filtering
Research: Experimental design, algorithm benchmarking, model validation, technical writing, data visualization